JACOB COOK

Junior Vulnerability Automation Engineer | Python · Web Scraping · Data Pipelines · Vulnerability Intelligence · PostgreSQL · Git · MITRE ATT&CK

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Python-focused security engineer with hands-on experience building data pipelines, parsing messy structured/unstructured sources, and automating detection content. Built Wazuh SIEM from scratch with FastAPI-based SOAR orchestration, Sigma-format rule authoring, and pytest-backed validation. Comfortable with regex, REST APIs, Git workflows, and turning research into reusable, scalable code. CompTIA stack (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress.

TECHNICAL SKILLS

Python & Data Engineering: Python (object-oriented), Web Scraping (HTML/JSON), Data Parsing & Transformation, Pipeline Orchestration (Apache Airflow familiar), API Implementation, REST APIs, FastAPI, Pytest-Backed Validation
Vulnerability Intelligence & Research: Vendor Advisory Analysis, CVE Research, Vulnerability Reports, Bug Trackers, Exploit Sources, OSINT, IOC Enrichment, Threat Intelligence Feeds, MITRE ATT&CK Familiarity
Databases & Tooling: PostgreSQL Familiarity, SQL, Git/GitHub Workflows, CI/CD Concepts (GitHub Actions–compatible), Linux Command Line, Regex, JSON/XML Parsing
Security Foundations: SIEM (Wazuh hands-on), Detection-as-Code, Sigma-Format Rules, Phishing/IOC Analysis, Wireshark, Azure AD, Okta, Cloud (AWS labs)

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code 2026
- Authored 6 Sigma-format detection rules for Intune MDM abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation (80 benign suppressed, 20 malicious caught) using KV store-style admin baseline lookups.
- Documented 3 benign false-positive scenarios per rule (18 total), covering VPN logins, scheduled maintenance, and PIM activations so analysts can triage confidently without escalating known-good activity.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py), reproducible against synthetic data, not a live tenant.

Blue Team SOC Monitoring Lab: Wazuh SIEM 2026
- Built a full Wazuh SIEM stack from scratch and validated 12 detection rule patterns against 242 simulated auth-style events, separating true-positive threats from benign noise with 100% precision after tuning.
- Reduced false-positive alert volume by 100% by tuning out all 79 benign-only matches (cron jobs, service accounts, scheduled deployments) without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py). All metrics reproducible.

SOAR-lite: Incident Response Orchestrator 2026
- Eliminated 4 manual triage steps (classification, IOC enrichment, notification, ticket creation) by building automated SIEM-triggered playbooks, reducing triage time by over 60% in lab testing.
- Built end-to-end alert automation: webhook ingestion from Wazuh, automatic playbook routing by event type, IP blocking, Slack notification, and incident documentation with zero analyst intervention required.
- Delivered a working handoff path from initial alert handling to documented, Tier 2-ready response that closes the gap between raw detection and actionable cases.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against the real FastAPI app, not mocked endpoints.

Phishing Analysis Lab 2026
- Triaged 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction, and attacker infrastructure mapping, achieving 100% accurate threat classification against known IOC ground truth.
- Delivered production-style triage reports with documented IOCs, confidence assessments, and escalation decisions that mirror real security operations output.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6) 2024
- Secured a production AI platform for corporate stakeholders by engineering RBAC and Azure AD SSO, enforcing data privacy requirements that enabled executive-level adoption.
- Shipped an enterprise-grade platform on deadline as part of a six-person team under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response 2019 - 2025
- Maintained 99.5% system uptime across automated sortation and conveyor infrastructure by monitoring sensors, alerts, and operational dashboards across rotating shifts, the same continuous monitoring discipline that defines round-the-clock SOC work.
- Resolved 200+ system anomalies over six years through structured root cause analysis, producing compliance-tracked documentation that reduced repeat incidents by an estimated 30% and supported audit readiness.
- Reduced mean time to recovery by an estimated 20% by coordinating cross-team incident response during outages, including escalation, triage prioritization, and post-incident reporting that mirrors security incident response workflows.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress) Expected Oct 2026 · GPA 3.50
CSUS | BS, Computer Science Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)